Activity 1
Mean Salary

Setting of the problem

We are given the following information:

> A random sample of the salaries assigned to the employees of a given company showed the following data distribution:
>
> 1. 7,000 to each of the 50 operating employees
> 2. 25,000 to each of the 35 trusted employees
> 3. 60,000 to each of the 16 area managers
> 4. 100,000 to the general manager

We are then asked to answer the following questions:

> a) What is the mean, mode, median and range of the company?
> b) Which measure of centralization is better to represent the salaries of the company employees?
> c) What is the standard deviation and variance?
> d) Compute what percentage of the given data is within one standard deviation around the mean

In this report we will show how to obtain the answer to these questions using Quarto.

Computing the required data using R

We start by creating a lookup table relating the employee codes with their types for easy reference:

```r
library(tidyverse)
library(krulRutils)
options(scipen = 999) # Disable scientific notation


# Employee codes and types
employee_code_vector <- c("oper_emp", "trust_emp", "area_mgr", "gen_mgr")

employee_type_vector <- c(
  "Operating Employee",
  "Trusted Employee",
  "Area Manager",
  "General Manager"
)

# Employee lookup tibble
employee_lookup_tbl <- tibble(
  employee_code = employee_code_vector,
  employee_type = employee_type_vector
)
```

We now create a table containing the employee salaries:

```r
# Employee salary data
employee_salary_vector <- c(7000, 25000, 60000, 100000)

# Employee frequency vector
employee_frequency_vector <- c(50, 35, 16, 1)

# Employee salary tibble
employee_salary_tbl <- tibble(
  employee_code = rep(employee_code_vector, employee_frequency_vector),
  employee_salary = rep(employee_salary_vector, employee_frequency_vector)
)
```

Using this table, we compute the mean, median and range:

```r
salaries = employee_salary_tbl$employee_salary
salaries_mean = mean(salaries)
salaries_median = median(salaries)
salaries_range = range(salaries)
```

To compute the mode, we first need to define the corresponding function and then we apply it to our data vector:

```
mode <- function(x) {
   return(as.numeric(names(which.max(table(x)))))
}
salaries_mode=mode(salaries)
```

We now proceed to compute the variance and the standard deviation:

```
salaries_sd=sd(salaries)
salaries_var=var(salaries)
```

To find the percentage of data within one standard deviation of the mean, we first build a data vector containing only those elements satisfying the aforementioned condition:

```
salaries_within_sd=salaries[abs(salaries - salaries_mean) <= salaries_sd]
```

After that, we simply compare the lenghts of the corresponding vectors:

```
percentage_within_sd=length(salaries_within_sd)/length(salaries)*100
```

We will also want to graph this information, we start by assigning colors to our employee salaries:

```
#Employee salary colors
employee_salary_colors <- c(
  "#1f77b4", # Operating employee
  "#2ca02c", # Trusted employee
  "#ff7f0e", # Area manager
  "#d62728"  # General manager
)
```

We now create a table with the employee types, salaries and their frequencies:

```
employee_salary_freq_tbl <- employee_salary_tbl %>%
  count(employee_code, employee_salary, name = "frequency", .drop = FALSE) %>%
  arrange(employee_salary) %>%
  convert_codes_to_factor(
    code_col = employee_code,
```

```r
    lookup_tbl = employee_lookup_tbl,
    lookup_code_col = employee_code,
    lookup_label_col = employee_type,
    label_col = employee_type
  ) %>%
  mutate(employee_salary = factor(
    employee_salary,
    levels = employee_salary_vector,
  ))
```

Using this table, we can generate the corresponding plot:

```r
# Plot employee salary frequency
employee_salary_freq_plot <- ggplot(
  employee_salary_freq_tbl,
  aes(
    x = employee_type,
    y = frequency,
    fill = employee_salary,
    label = frequency
  )
) +
  geom_col(
    color = "black",
    position = position_dodge(width = 0.9)
  ) +
  geom_text(
    stat = "identity",
    vjust = -0.5,
    #position = position_stack(vjust = 1)
    position = position_dodge(width = 0.9)
  ) +
  scale_fill_manual(values = employee_salary_colors) +
  labs(
    title = "Employee Salary Frequency Plot",
    x = "Employee Types",
    y = "Frequency",
    fill = "Employee Salaries"
  )
```

With all this information at hand, we can now proceed to answer the questions given in this activity.

Solutions to the given questions

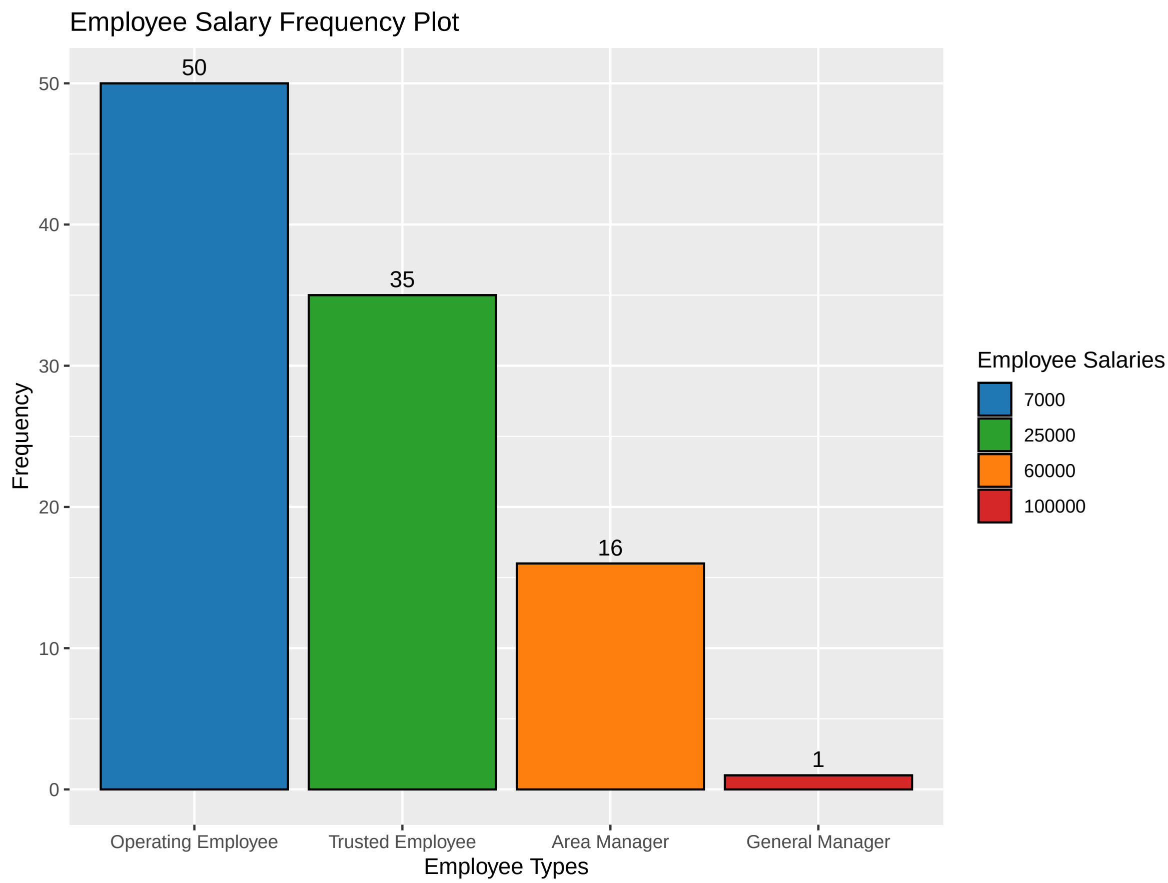
a) What is the mean, mode, median and range of the company?

The values that we are looking for are:

Mean = 22401.9607843
Mode = 7000
Median = 25000
Range = 7000, 100000

b) Which measure of centralization is better to represent the salaries of the company employees

In this case our data is very skewed as it can be seen from the following graph:



Therefore it is recommended that we use the *median* as our measure or centralization. Notice, however that the mean and the median are relatively close in this case.

c) What is the standard deviation and variance?

The values required are:

Standard Deviation = 20083.1241919
Variance= 403331877.305378

‘

d) Compute what percentage of the given data is within one standard deviation around the mean

The percentage of the salaries that are within one standard deviation of the mean is: 83.3333333%